

Lecture 04

BENZENE AND AROMATICITY

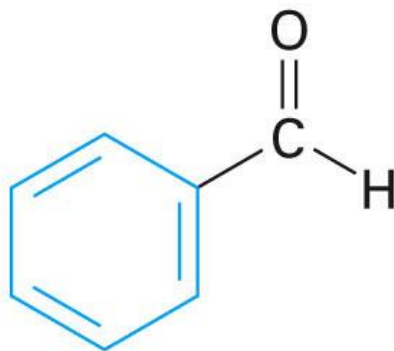
Aromatic Compounds

- *Aromatic* was used to describe some fragrant compounds in early 19th century
 - Not correct: later they are grouped by chemical behavior (unsaturated compounds that undergo substitution rather than addition)
- Current: distinguished from *aliphatic* compounds by electronic configuration

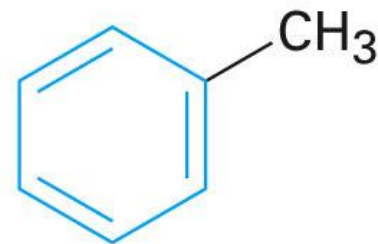


Benzene

© 2007 Thomson Higher Education



Benzaldehyde



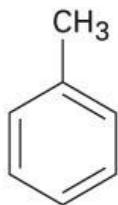
Toluene

Sources and Names of Aromatic Hydrocarbons

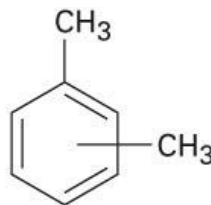
- From high temperature distillation of coal tar
- Heating petroleum at high temperature and pressure over a catalyst



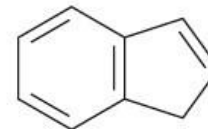
Benzene
(bp 80 °C)



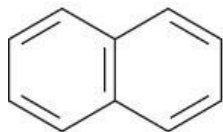
Toluene
(bp 111 °C)



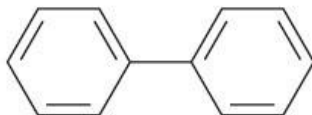
Xylene
(bp: ortho, 144 °C;
meta, 139 °C; para, 138 °C)



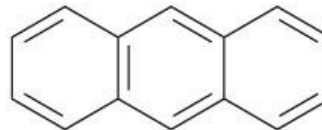
Indene
(bp 182 °C)



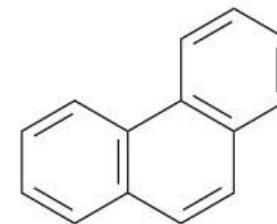
Naphthalene
(mp 80 °C)



Biphenyl
(mp 71 °C)



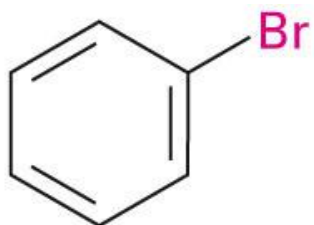
Anthracene
(mp 216 °C)



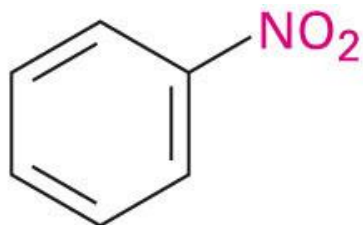
Phenanthrene
(mp 101 °C)

Some Aromatic Compounds

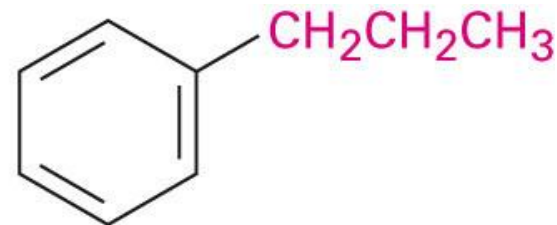
- Many common names (toluene = methylbenzene; aniline = aminobenzene)
- Monosubstituted benzenes systematic names as hydrocarbons with –*benzene*
 - $\text{C}_6\text{H}_5\text{Br}$ = bromobenzene
 - $\text{C}_6\text{H}_5\text{NO}_2$ = nitrobenzene, and $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_3$ is propylbenzene



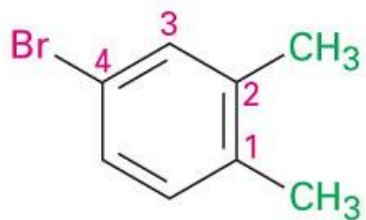
Bromobenzene



Nitrobenzene

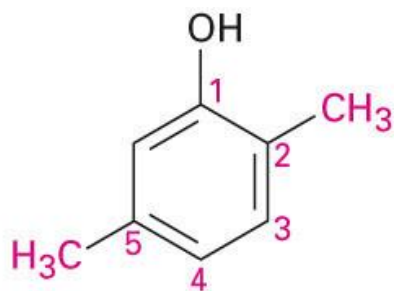


Propylbenzene



4-Bromo-1,2-dimethylbenzene

© 2007 Thomson Higher Education



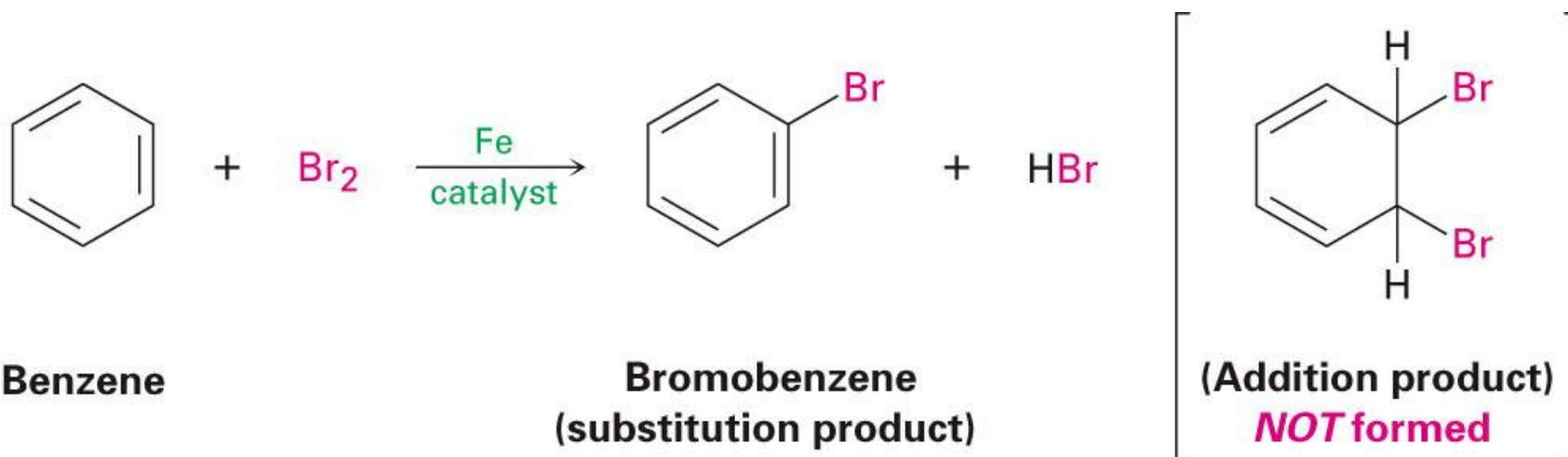
2,5-Dimethylphenol



2,4,6-Trinitrotoluene (TNT)

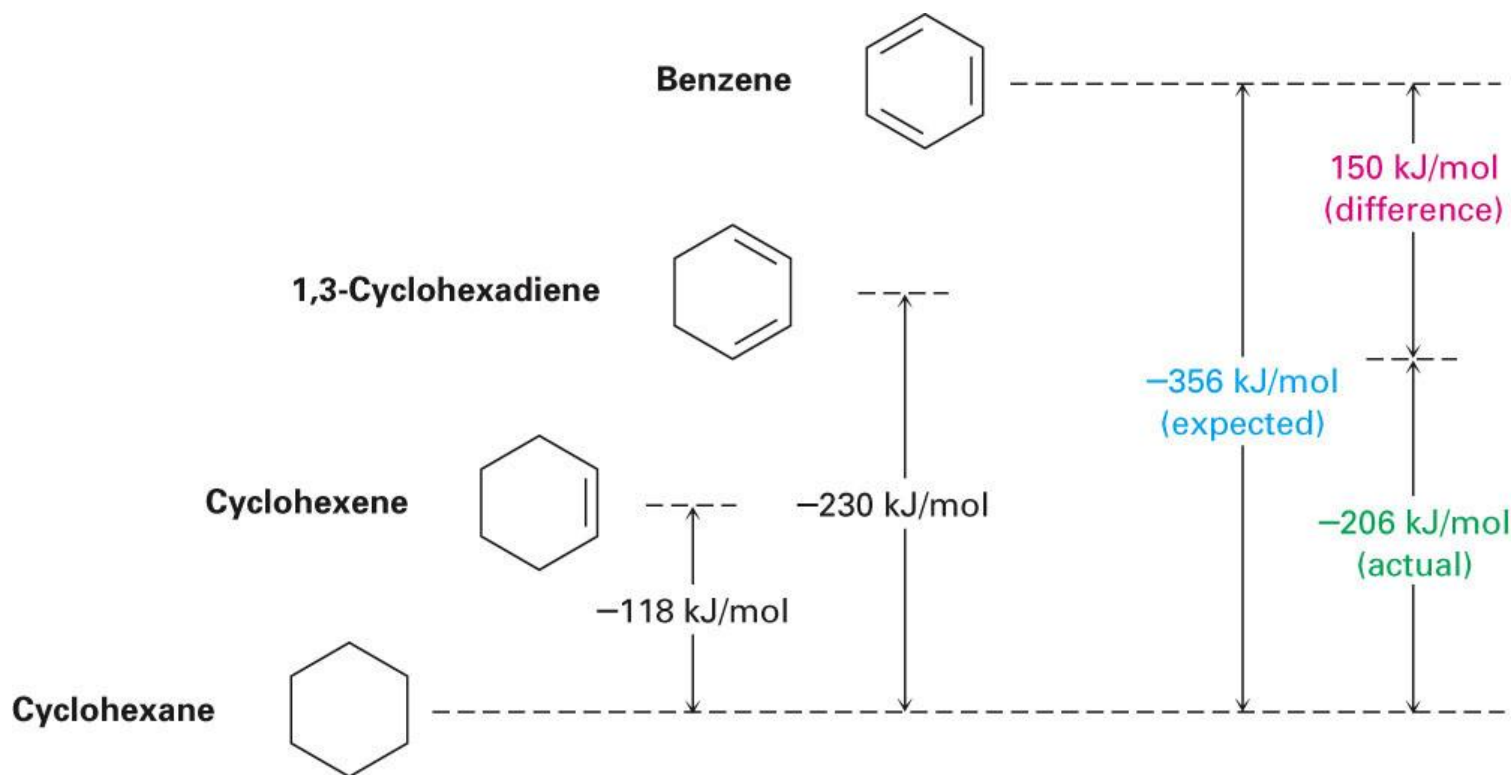
Structure and Stability of Benzene: Molecular Orbital Theory

- Benzene reacts slowly with Br_2 to give bromobenzene (where Br replaces H)
- This is substitution rather than the rapid addition reaction common to compounds with $\text{C}=\text{C}$, suggesting that in benzene there is a higher barrier

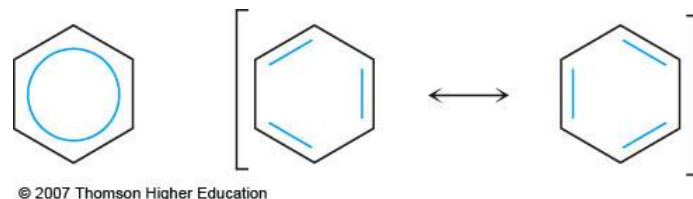
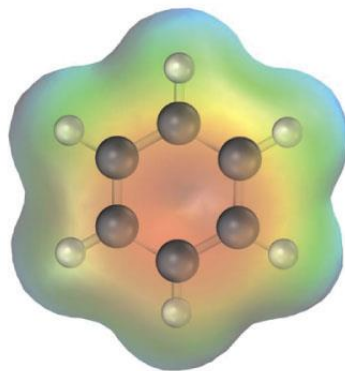
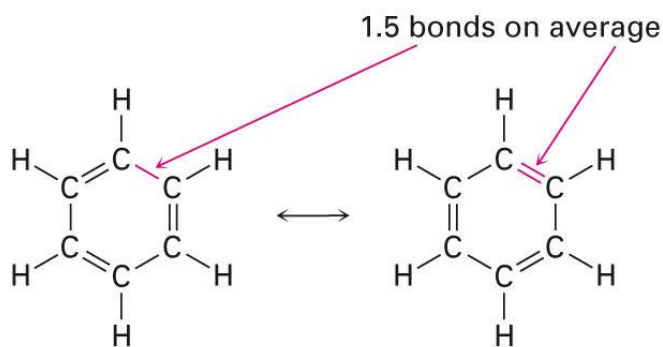


Heats of Hydrogenation as Indicators of Stability

- The addition of H_2 to $\text{C}=\text{C}$ normally gives off about 118 kJ/mol – 3 double bonds would give off 356 kJ/mol
 - Two conjugated double bonds in cyclohexadiene add 2 H_2 to give off 230 kJ/mol
- Benzene has 3 unsaturation sites but gives off only 206 kJ/mol on reacting with 3 H_2 molecules
- Therefore it has about 150 kJ more “stability” than an isolated set of three double bonds



- All its C-C bonds are the same length: 139 pm — between single (154 pm) and double (134 pm) bonds
- Electron density in all six C-C bonds is identical
- Structure is planar, hexagonal
- C–C–C bond angles 120°
- Each C is sp^2 and has a p orbital perpendicular to the plane of the six-membered ring



Aromaticity and the Hückel $4n+2$ Rule

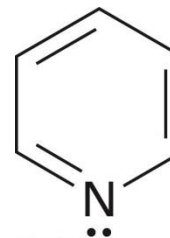
- Unusually stable - heat of hydrogenation 150 kJ/mol less negative than a cyclic triene
- Planar hexagon: bond angles are 120° , carbon-carbon bond lengths 139 pm
- Undergoes substitution rather than electrophilic addition
- Resonance hybrid with structure between two line-bond structures



© 2007 Thomson Higher Education

Benzene

Three double bonds;
six π electrons



© 2007 Thomson Higher Education

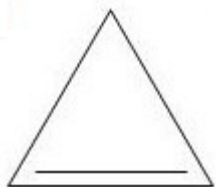
Pyridine

- **Huckel's rule for Aromaticity**

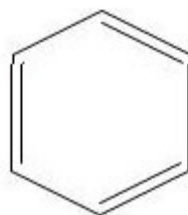
A planar cyclic molecule with alternating double and single bonds has aromatic stability if it has $4n+2$ π electrons (n is 0,1,2,3,4 – any integer)

For $n=1$: $4n+2 = 6$; benzene is stable and the electrons are delocalized

- Compounds with $4n$ π electrons are not aromatic (May be Antiaromatic)
 - Planar, cyclic molecules with **$4n$** π electrons are much *less* stable than expected (antiaromatic)
 - They will distort out of plane and behave like ordinary alkenes
 - 4- and 8-electron compounds are not delocalized (single and double bonds)



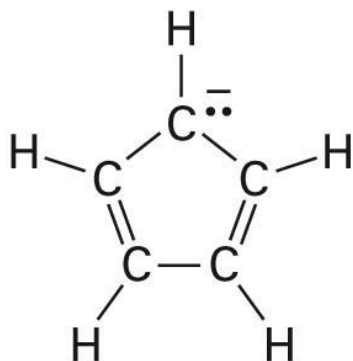
non aromatic



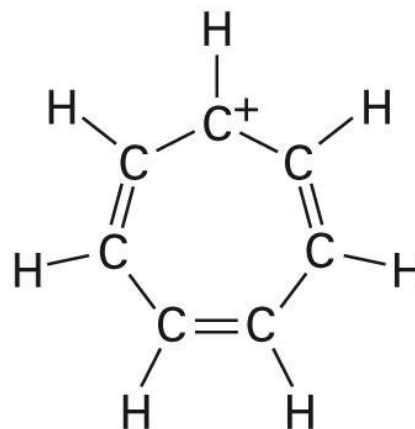
aromatic

Aromatic Ions

- The $4n + 2$ rule applies to ions as well as neutral species
- Both the cyclopentadienyl *anion* and the cycloheptatrienyl *cation* are aromatic
- The key feature of both is that they contain 6π electrons in a ring of continuous p orbitals



Cyclopentadienyl anion

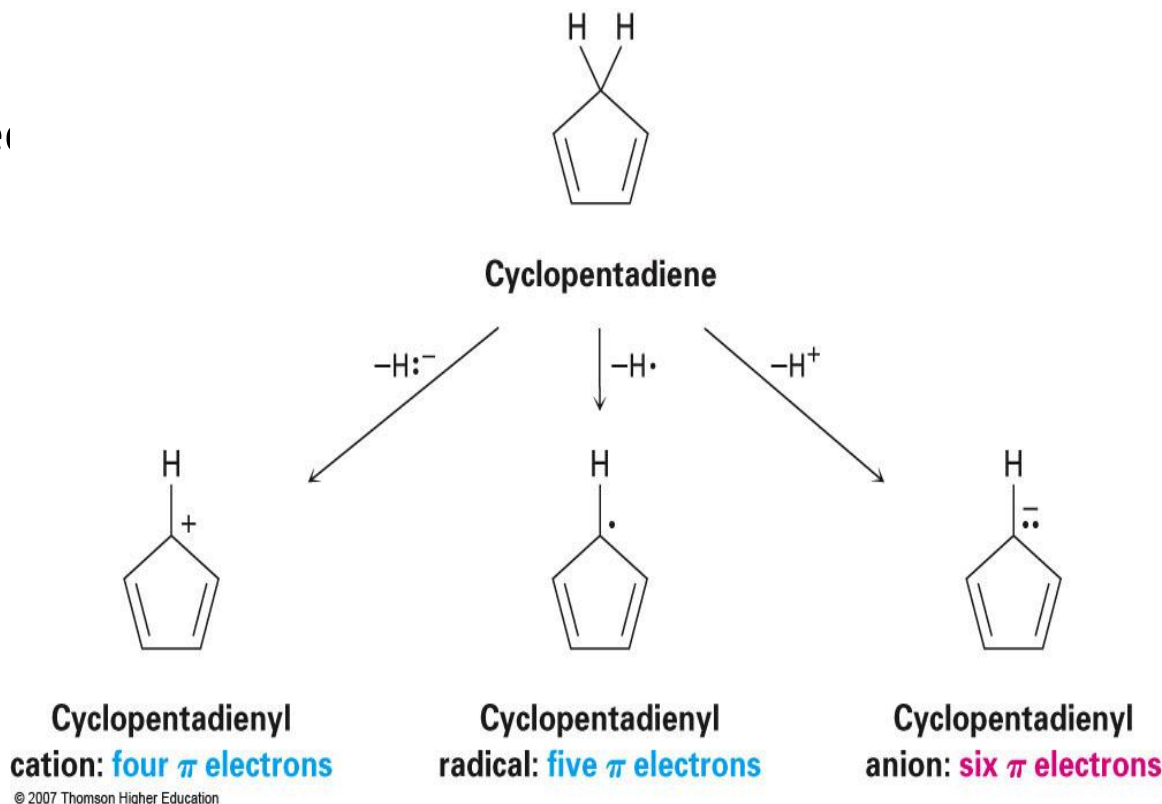


Cycloheptatrienyl cation

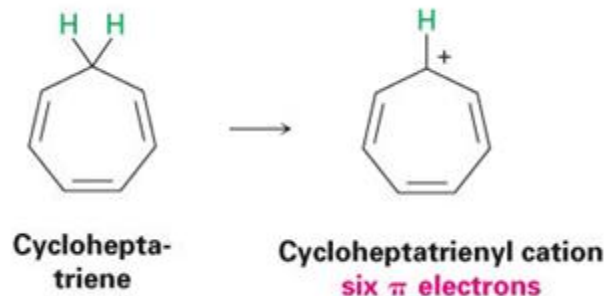
Six π electrons; aromatic ions

Aromaticity of the Cyclopentadienyl Anion

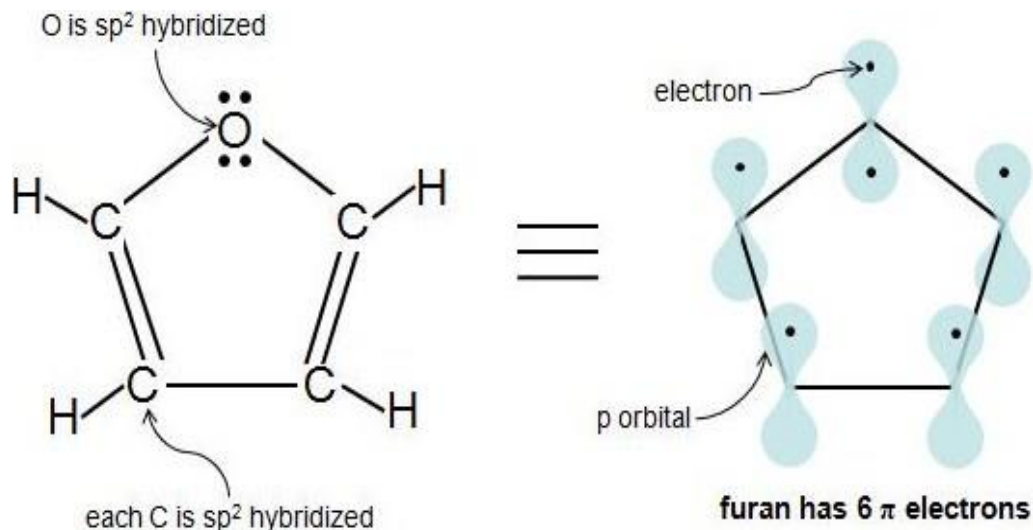
- 1,3-Cyclopentadiene contains conjugated double bonds joined by a CH_2 that blocks delocalization
- Removal of H^+ at the CH_2 produces a cyclic 6-electron system, which is stable
- Removal of H^- or $\text{H}^\bullet$ generates nonaromatic 4 and 5 electron systems
- Relatively acidic ($\text{pK}_\text{a} = 16$) because the anion is stable



- Cycloheptatriene has 3 conjugated double bonds joined by a CH_2
- Removal of "H" leaves the cation
- The cation has 6 electrons and is aromatic



- compounds with elements other than carbon in the ring can also be aromatic, as long as they fulfill the criteria for aromaticity. Eg N,S,O,P (Furan, Pyridine- $\text{C}_5\text{H}_5\text{N}$, Imidazole- $\text{C}_3\text{N}_2\text{H}_4$, Thiophene- $\text{C}_4\text{H}_4\text{S}$)



only 1 of oxygen's 2 lone electron pairs are π electrons (plane of the double bond). Furan has 6 π electrons and fulfills the $4n+2$ rule

Arenes

- Derivatives of Benzene (Eg: substituent : Alkyl group)
- Alkyl polynuclear systems (Naphthalene, Anthracene...)
- Arene Synthesis:
 - Two methods
 - Alkylation (Friedel-Crafts reaction)
 - Acylation

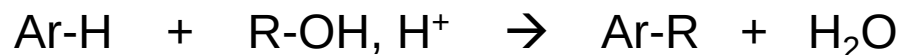
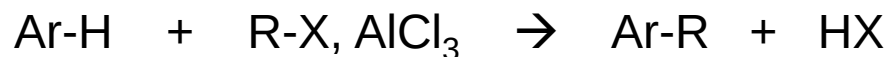
Friedel-Crafts reaction

Catalysts:

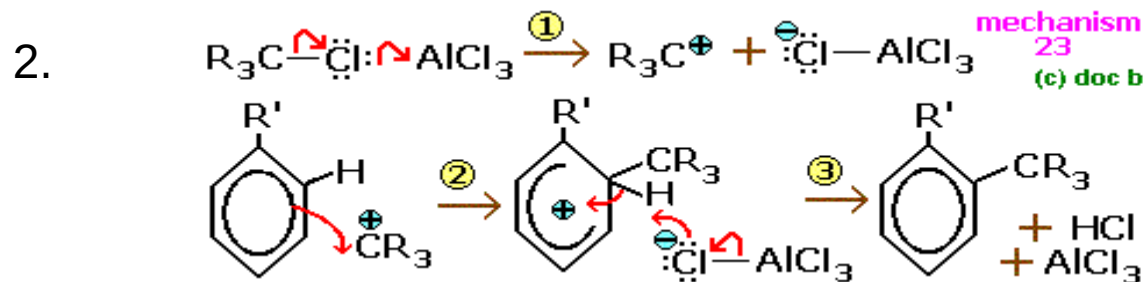
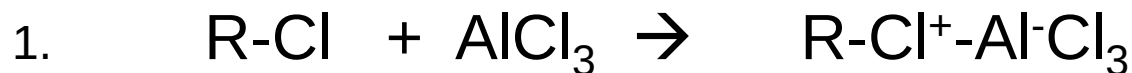
Metal Halides (Lews acids) $\text{AlCl}_3 > \text{BF}_3 > \text{SbCl}_3 > \text{FeCl}_3 > \text{SnCl}_3 > \text{ZnCl}_3$

Proton acid catalysis: HF , H_2SO_4 , H_3PO_4

Friedel-Crafts:



- Reaction rate varies : $\text{tertiary-R} > \text{secondary -R} > \text{primary -R}$
 $\text{R-F} > \text{R-Cl} > \text{R-Br} > \text{RI}$
- Structure of the product depend on the structure of the alkyl group
- Mechanism for the Friedel-Craft alkylation

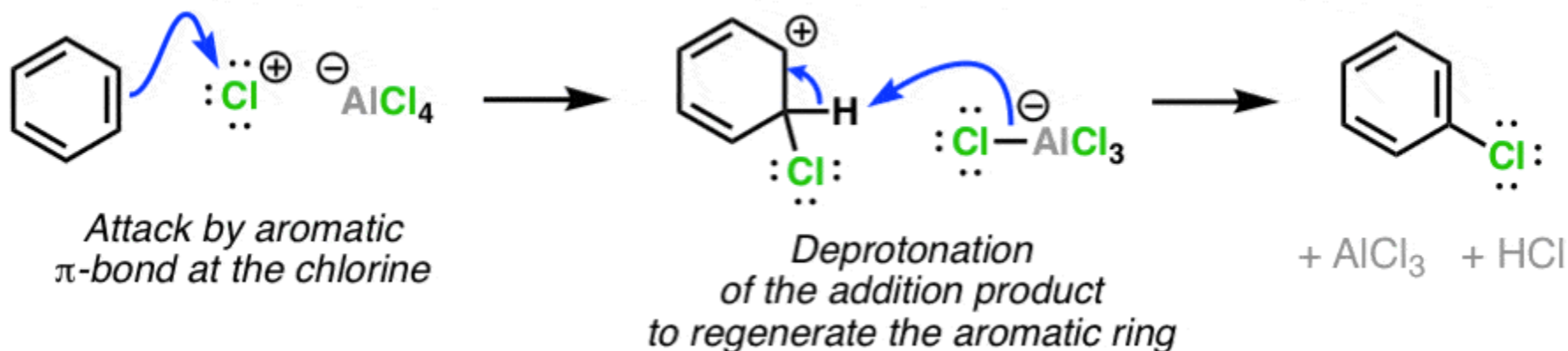


2.

How it works: Mechanism for activation of chlorine by AlCl_3



... and subsequent chlorination



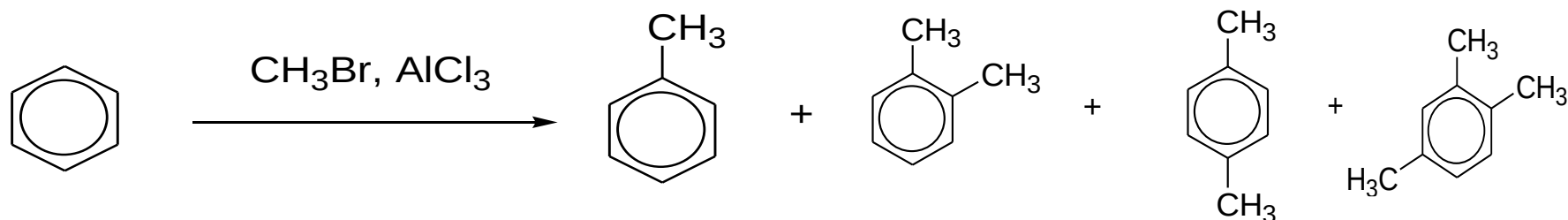
Friedel-Crafts limitations:

- a) Polyalkylation
- b) Possible rearrangement
- c) R-X cannot be Ar-X
- d) No reaction when the benzene ring is less reactive than bromobenzene
- e) No reaction with -NH_2 , -NHR , -NR_2 groups

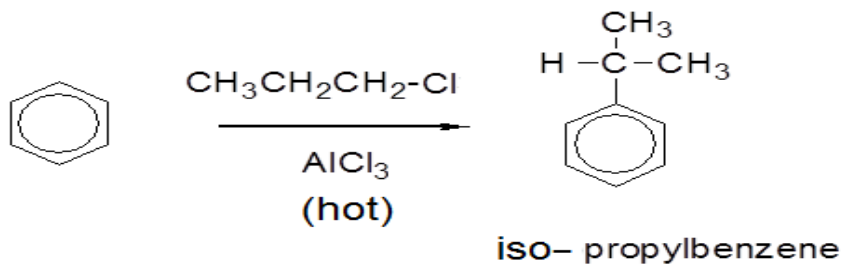
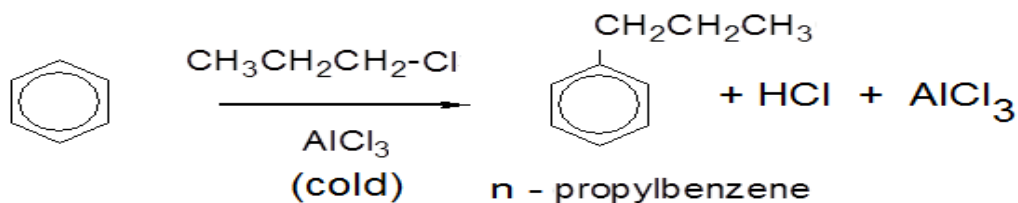
In syntheses it is often best to do Friedel-Crafts alkylation in the first step!

a) Polyalkylation

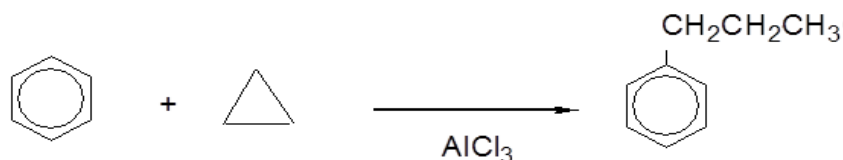
The alkyl group activates the ring making the products more reactive than the reactants leading to polyalkylation. Use of excess aromatic compound minimizes polyalkylation in the lab.



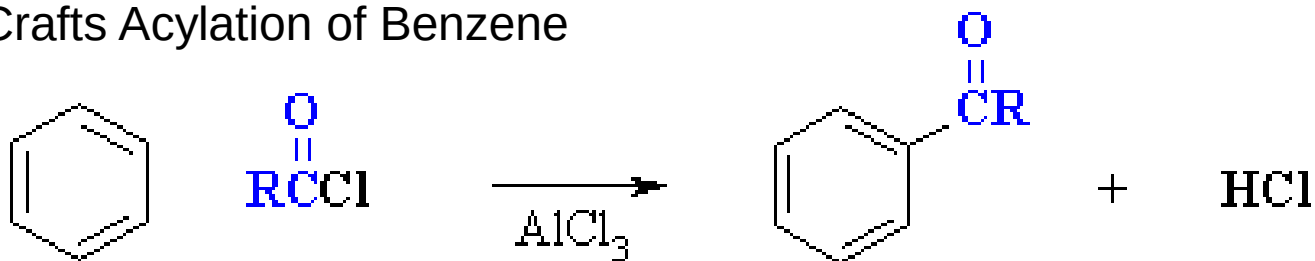
b) Rearrangement



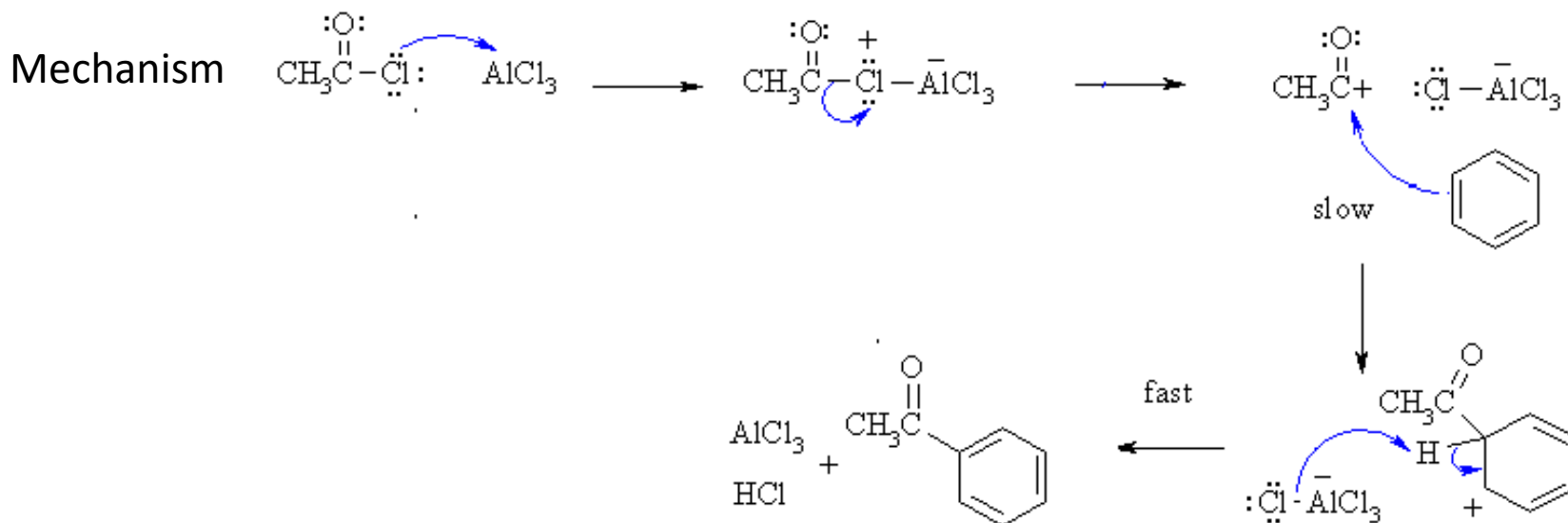
To avoid this rearrangement, cyclopropane can be used



Friedel-Crafts Acylation of Benzene

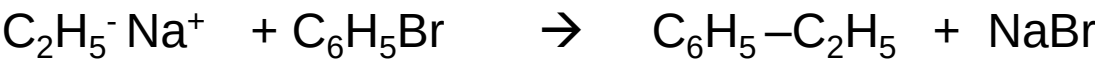
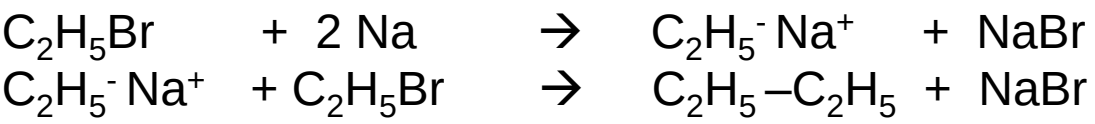
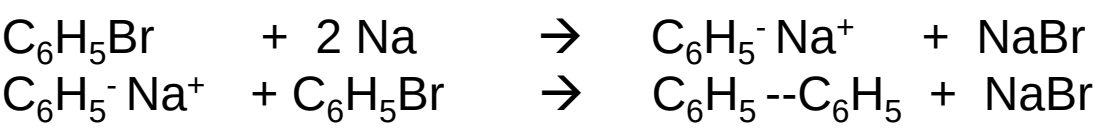
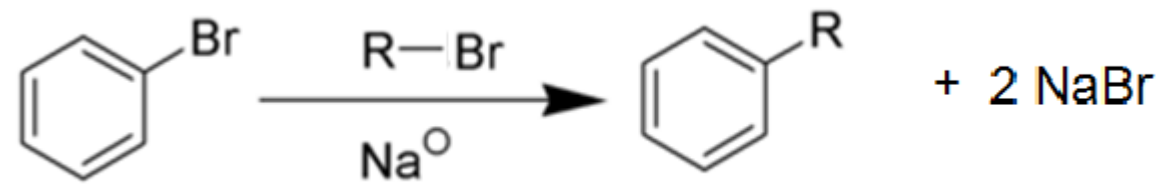


- Ar-H to Ar-COR(a ketone)
- Reagent : normally the acyl halide (e.g. usually RCOCl) with aluminum trichloride, AlCl_3 , a Lewis acid catalyst.
- Alternatively, the acid anhydride, i.e. $(\text{RCO})_2\text{O}$, can be used instead of the acyl halide.

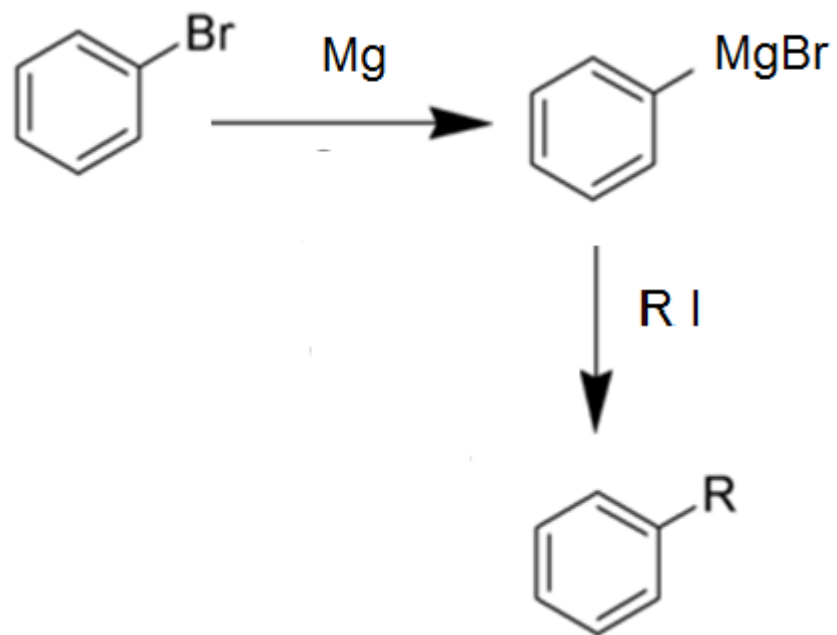


Wurtz-Fittig reaction :

Reaction of aryl halides with alkyl halides and sodium metal to give substituted aromatic compounds



Grignard reaction:



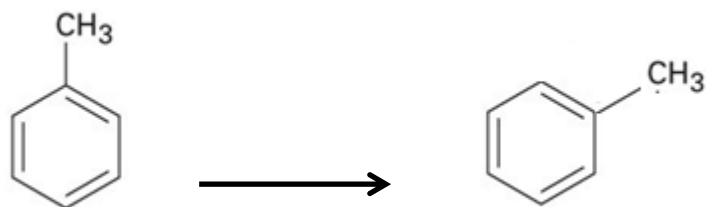
Properties of Arenes

Physical properties

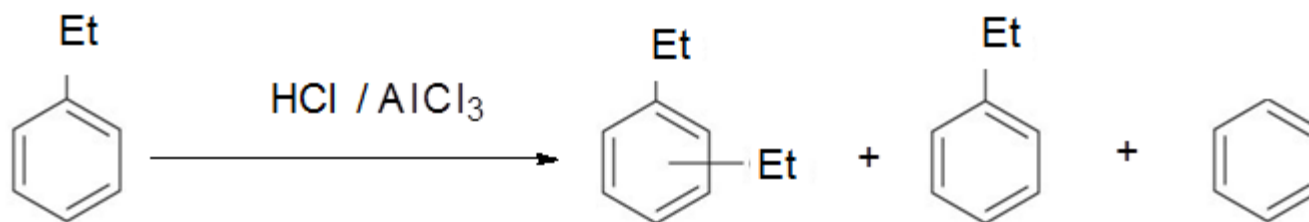
1. Colourless liquids
2. Boiling point regularly increases for each CH_2 group adding
3. BP of isomers are closer

Chemical properties

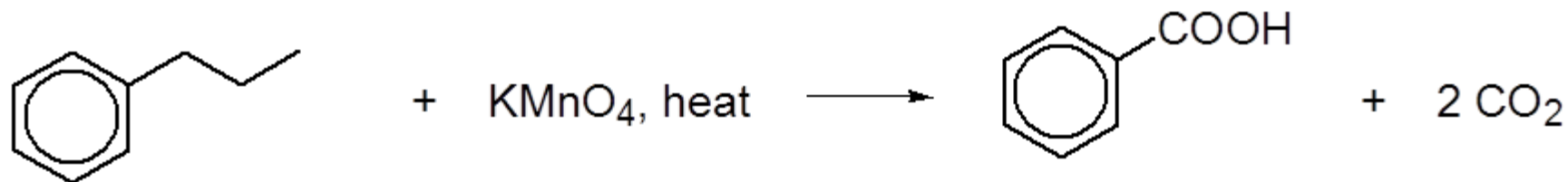
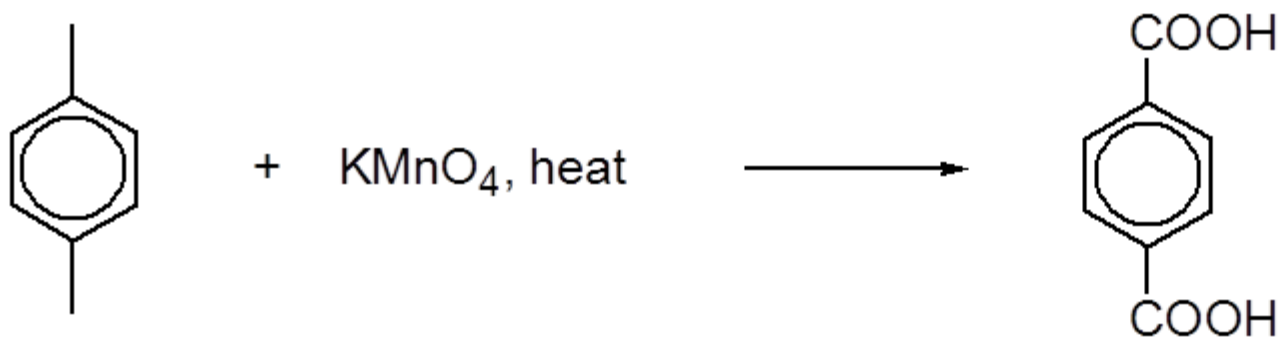
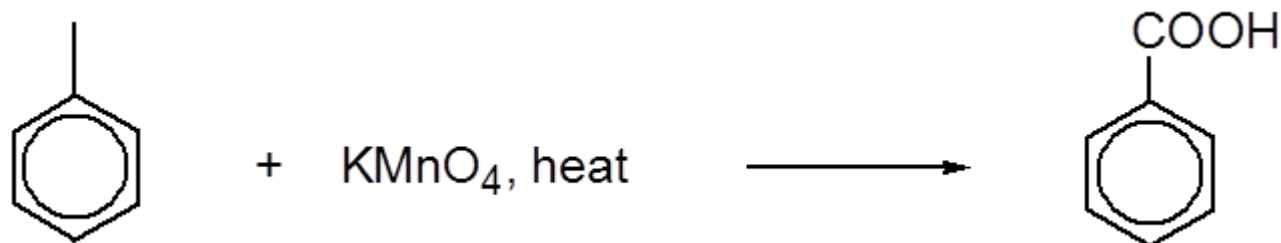
1. Rearrangement
2. Disproportionation -Isomers (monosubstituted, disubstituted.....)



Disproportionation -Isomers (monosubstituted, disubstituted)



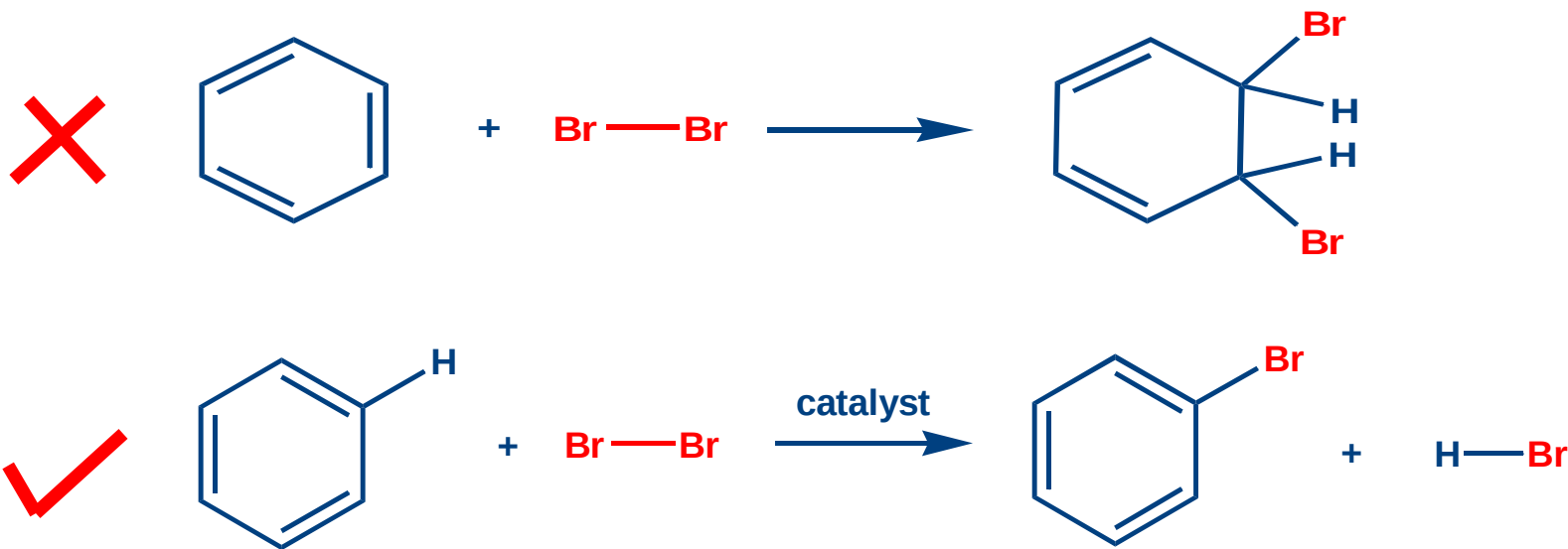
Oxidation of Arenes



Strong oxidizing agents : $\text{KMnO}_4 / \text{H}^+$, $\text{KMnO}_4 / \text{Alkane}$, $\text{Cr}_2\text{O}_7^{2-} / \text{H}^+$

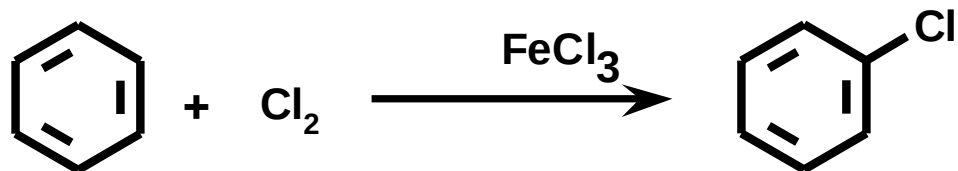
Aromatic Substitution Reaction

- Benzene does not undergo addition reactions to prevent the loss of its aromatic character.
- Undergoes substitution reactions by replacing a hydrogen atom of benzene (i.e., a substitution reaction), the aromatic character of the benzene ring is retained.
- This type of process remains energetically favorable.



Some Substitution Reactions of Benzene

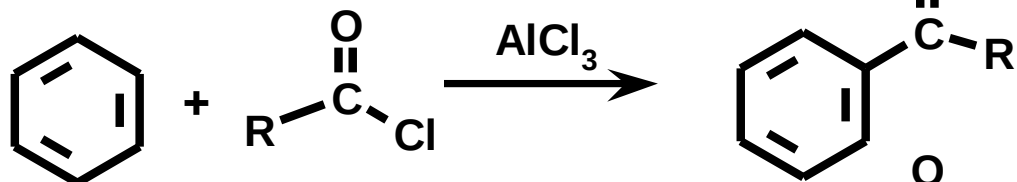
Halogenation



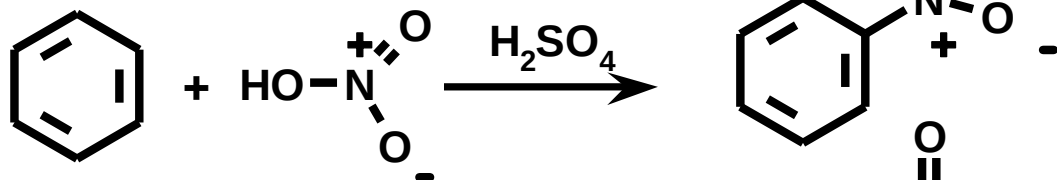
Friedel-Crafts Alkylation



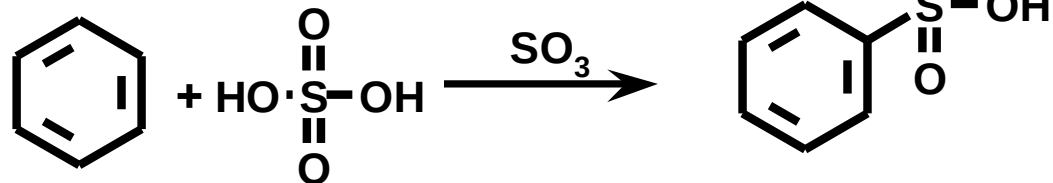
Friedel-Crafts Acylation



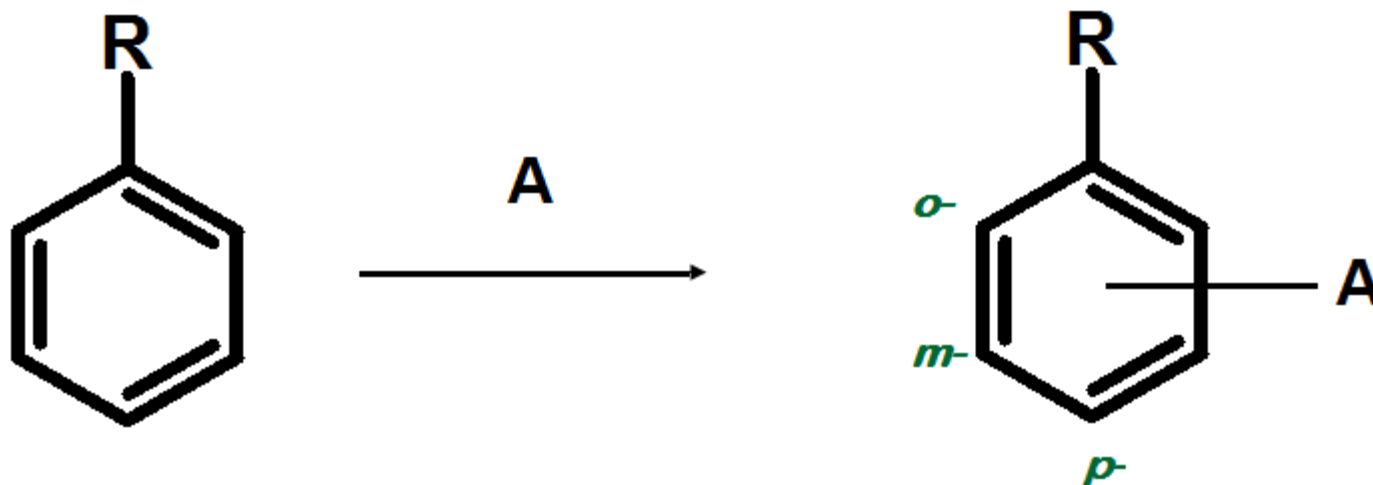
Nitration



Sulfonation



Second substitution



Substituent groups on a benzene ring affect electrophilic aromatic substitution reactions in two ways:

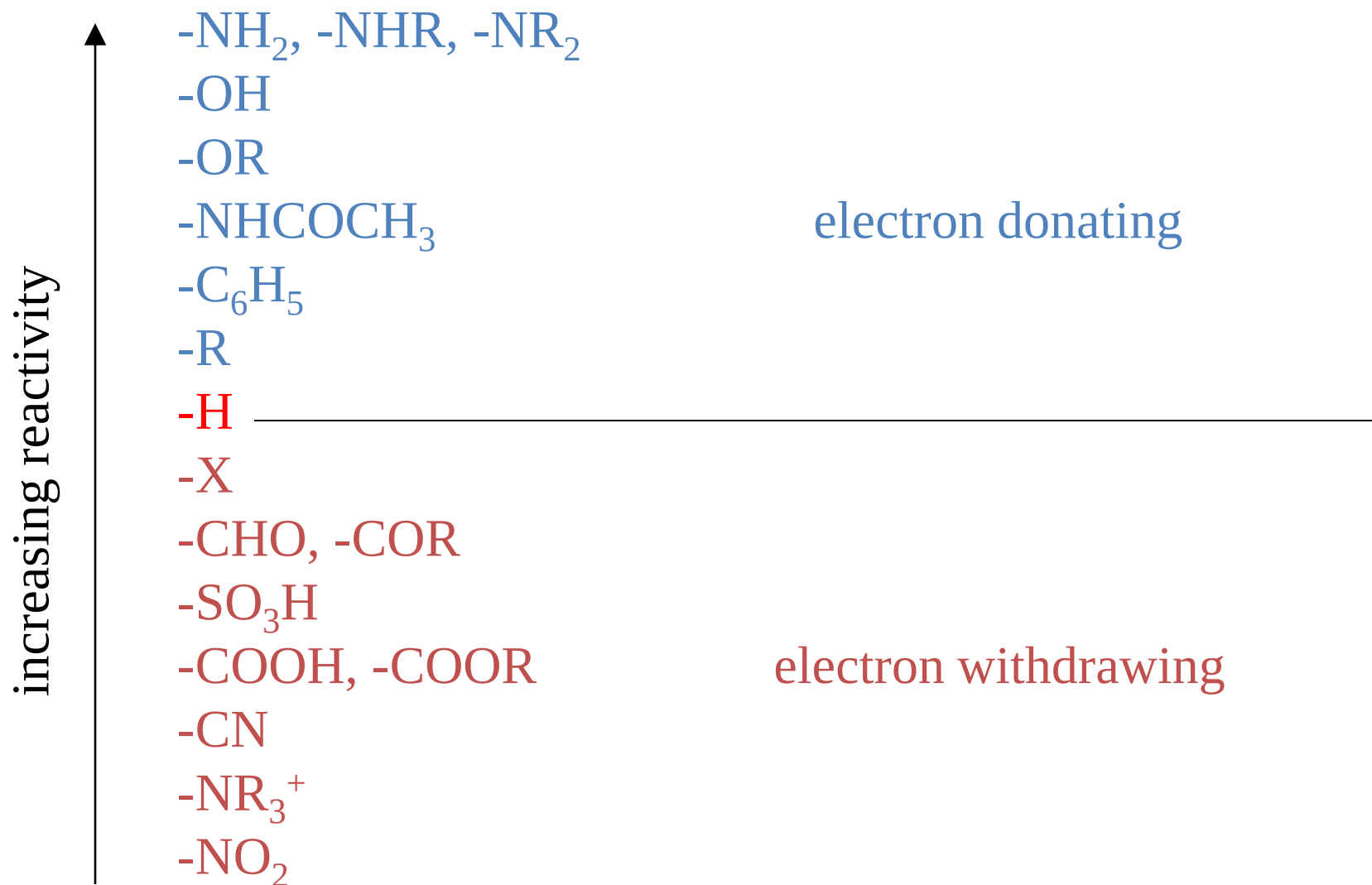
1) **reactivity**

activate (faster than benzene), or deactivate (slower than benzene)

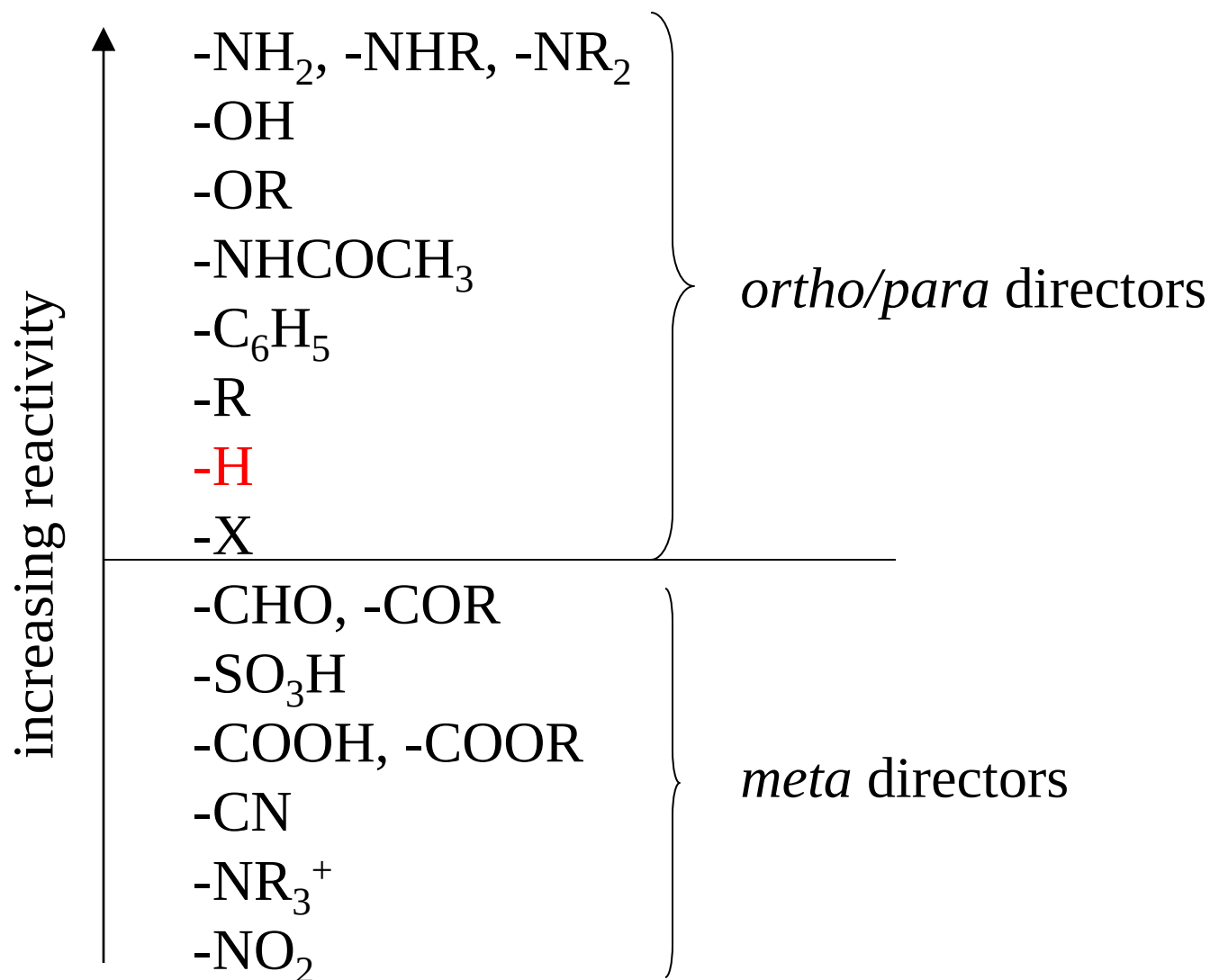
2) **orientation**

ortho- + *para-* direction, or *meta-* direction

Common substituent groups and their effect on **reactivity** in EAS:

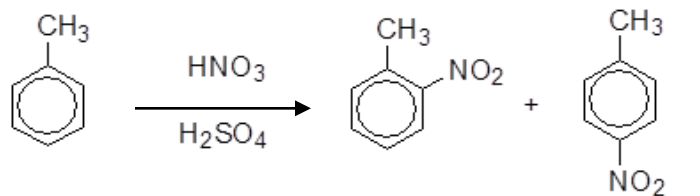


Common substituent groups and their effect on EAS:

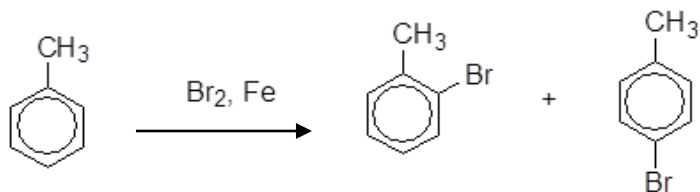


-CH₃ : activates the benzene ring towards EAS directs substitution to the *ortho*- & *para*- positions

Toluene

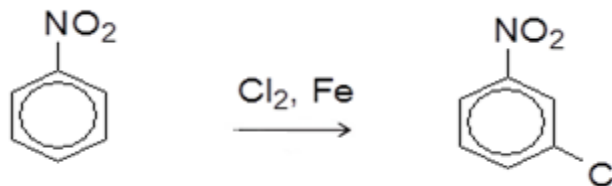


faster than the same reactions with benzene

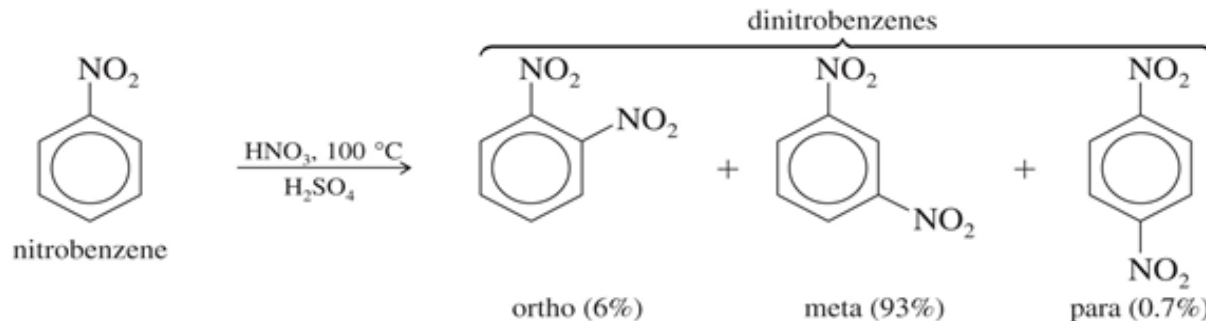


-NO₂ : deactivates the benzene ring towards EAS directs substitution to the *meta*-position

Nitrobenzene

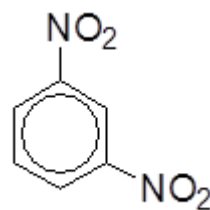
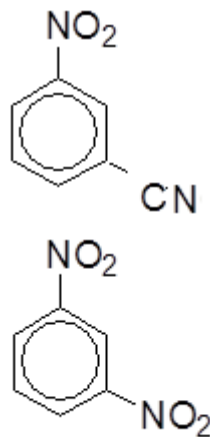
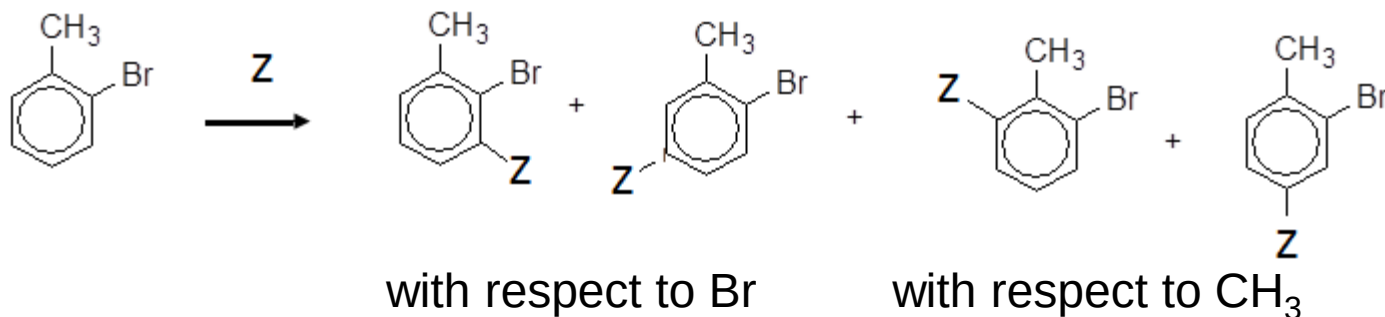
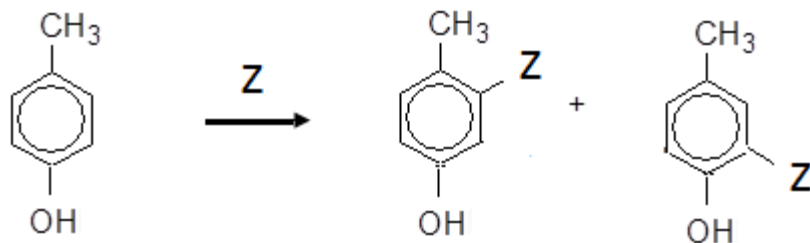


slower than the same reactions with benzene



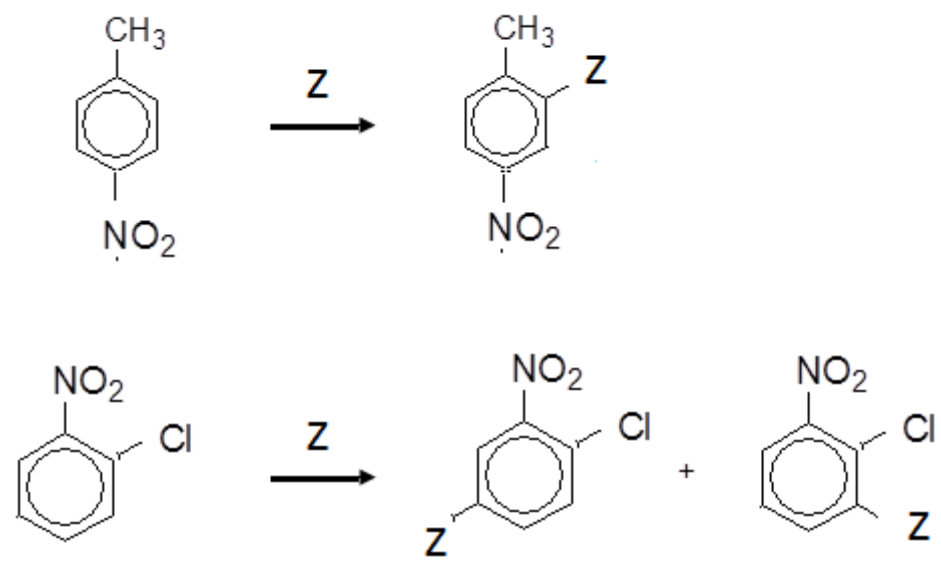
3rd substitution

Formation and the structure of the Tri substituted product depends on the structure of the 2nd substitution. When both group acts in same way, mixture of products can be obtained



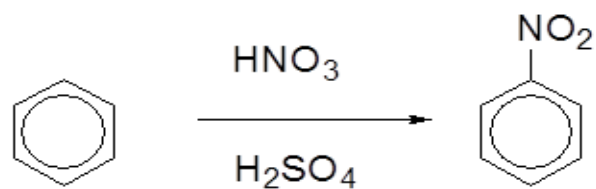
Both groups highly deactivate the ring,
thus 3rd substitution is not possible

When groups acts in different way, single product or mixture of products can be obtained

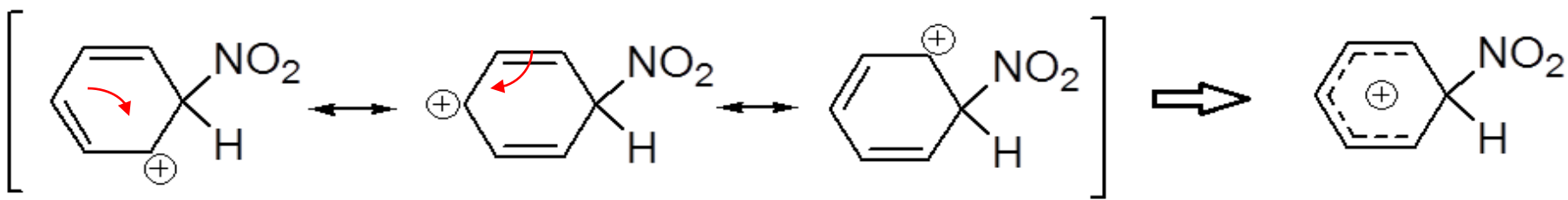
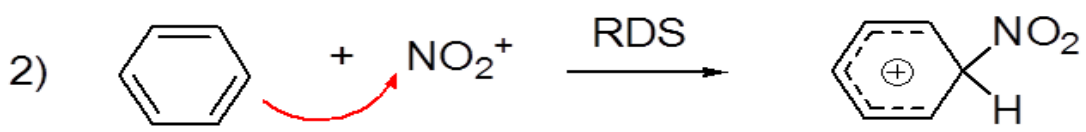


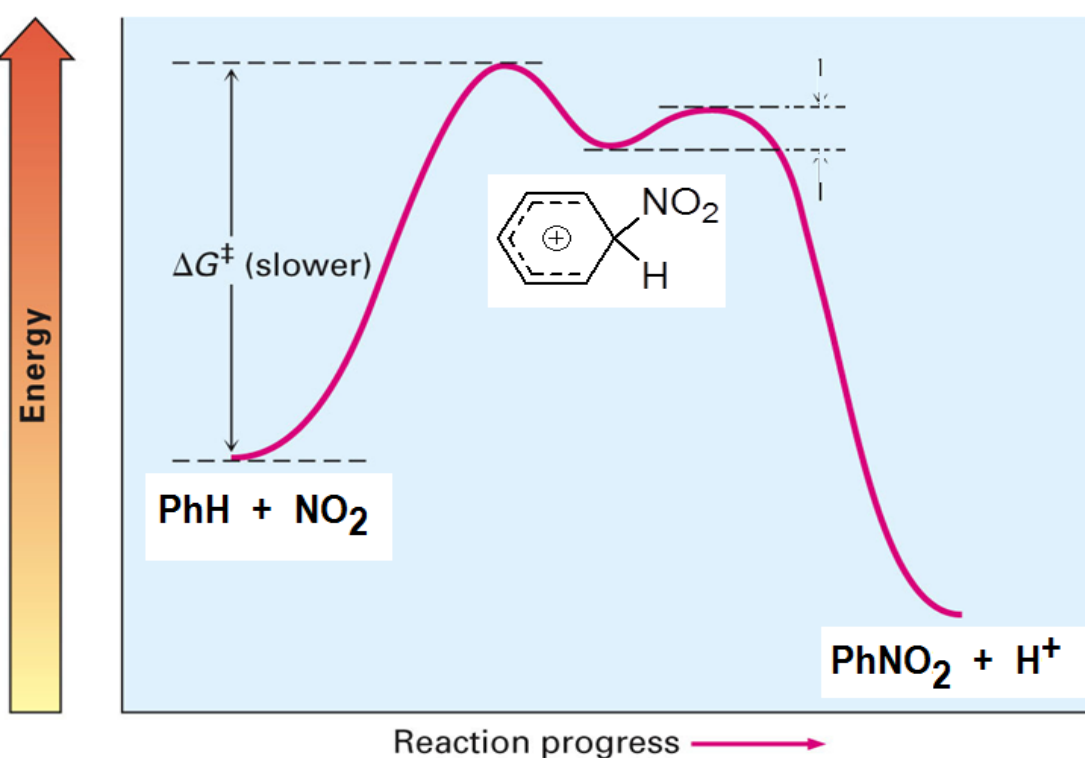
Aromatic nitro compounds

Nitration of benzene



Mechanism





Energy diagram

$$\text{Reaction rate} = K [\text{Substrate}] [\text{NO}_2^+]$$

- 2nd order

The reaction takes place by the arenium ion mechanism.

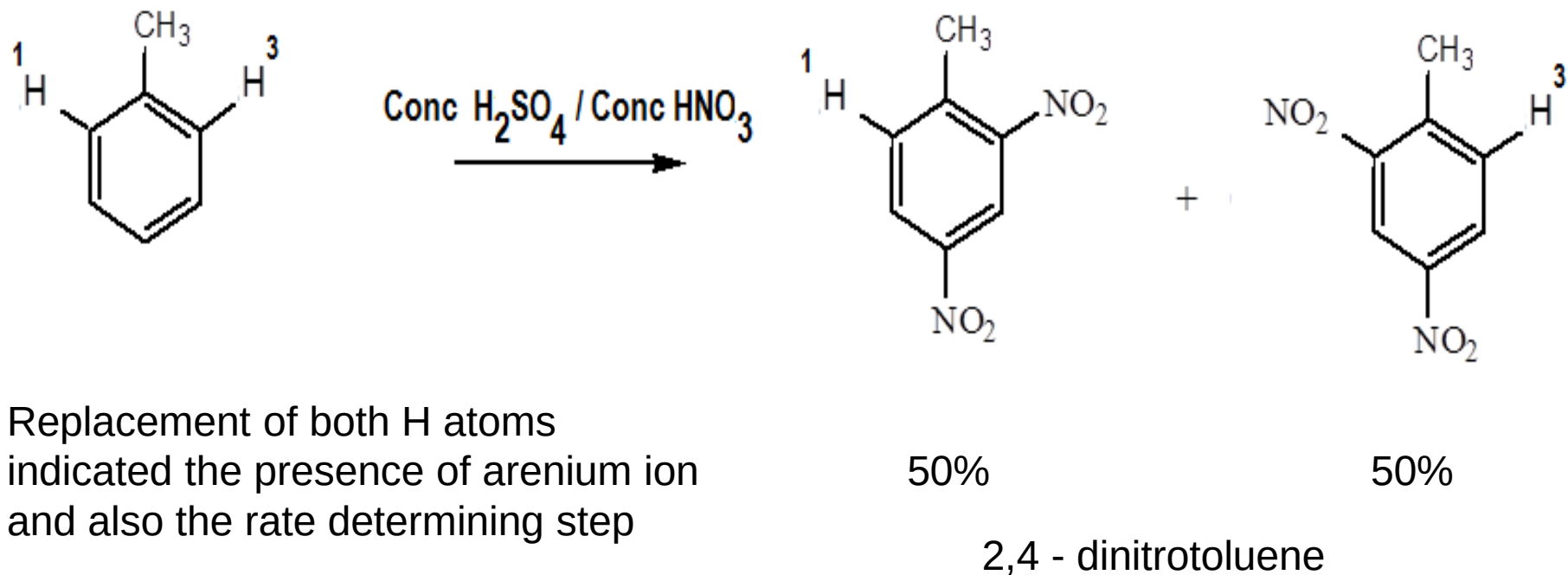
Step 1 (slow step):

Formation of a carbonium ion by the attack of the electrophilic on the aromatic substrate. The carbonium ion is stabilized by the resonance. This step is called arenium ion (sigma complex).

Step 2 (fast step):

The arenium ion loses a proton.

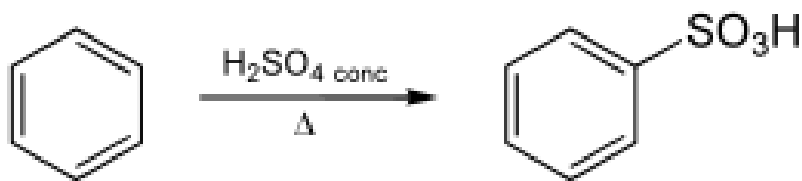
Evidence for arenium ion



Aromatic sulfonic Acids

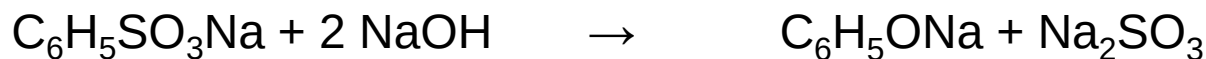
Benzenesulfonic acid is prepared from the sulfonation of benzene using concentrated sulfuric acid

This conversion illustrates aromatic sulfonation, which has been called "one of the most important reactions in industrial organic chemistry"

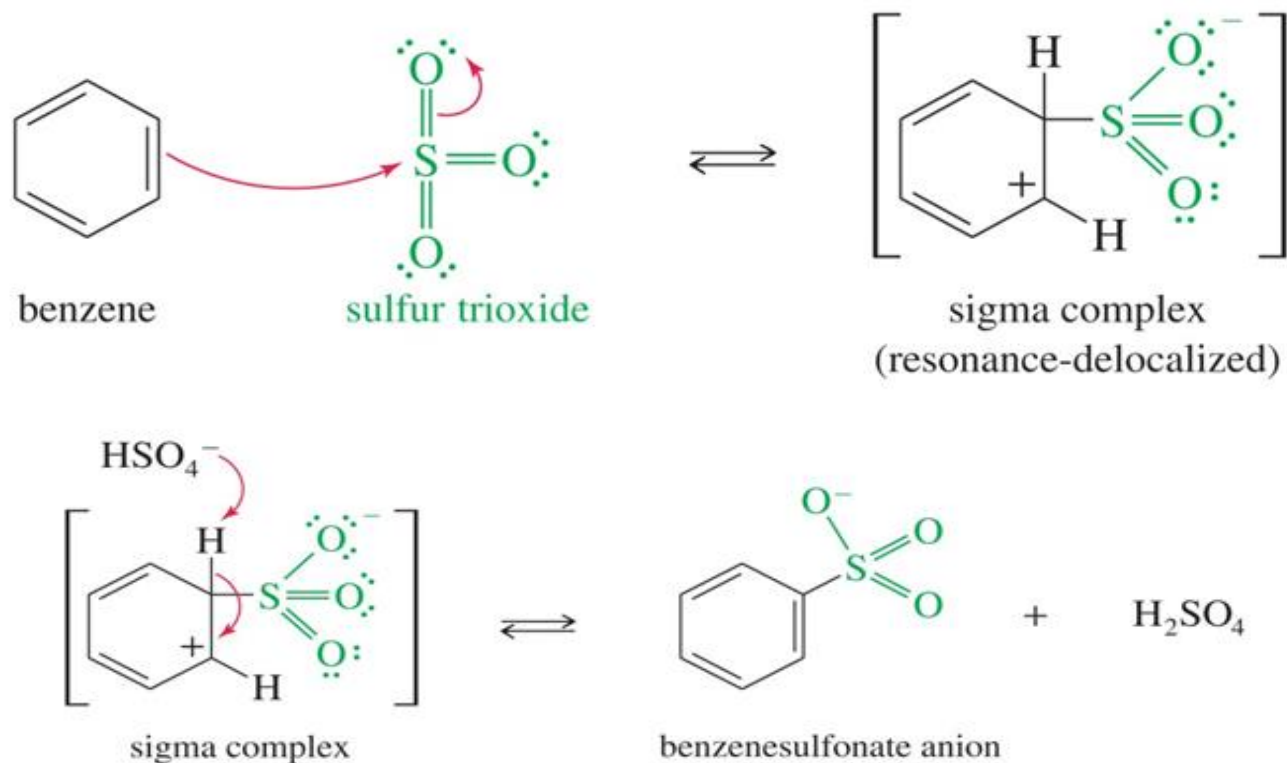


Applications:

The alkali metal salt of benzenesulfonic acid was once widely used in the production of phenol:

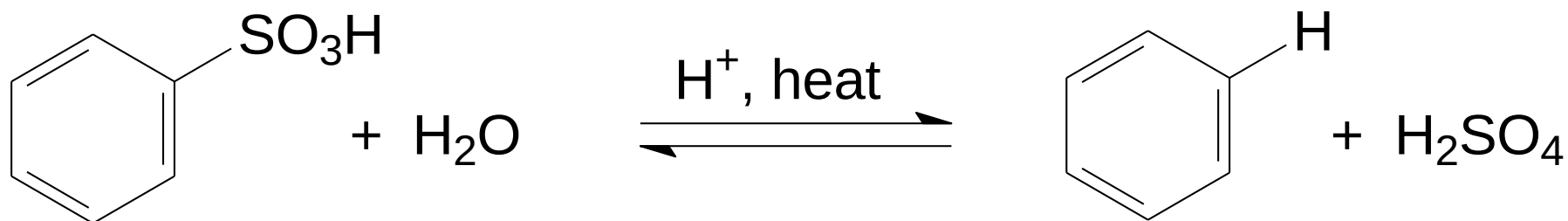


Mechanism of Sulfonation



- Benzene attacks sulfur trioxide, forming a sigma complex.
- Loss of a proton on the tetrahedral carbon and reprotonation of oxygen gives benzenesulfonic acid.

Desulfonation Reaction



- Sulfonation is reversible.
- The sulfonic acid group may be removed from an aromatic ring by heating in dilute sulfuric acid.